注意11

解：

∴原式 $=\int_0^1 dx\int_0^{1-x}dy\int_0^{1-x-y}e^{x+y+z}dz$

$=\int_0^1 dx\int_0^{1-x}(e-e^{x+y})dy$

$=\int_0^1(e^x-ex)=(e^x-\frac{1}{2}ex^2)\Big|_0^1=\frac{e}{2}-1.$

15. 取 $x=r\cos\theta$，$y=r\sin\theta$，$z=z$.

∴积分区域为 $r=z$ 与 $z=\pi$ 围成.

∴原式 $=\int_0^{\pi}r\,dr\int_0^{2\pi}d\theta\int_r^{\pi}\sin z\,dz$.

$=2\pi\int_0^{\pi}r(1+\cos r)dr=\pi^3-4\pi.$

16. 取 $x=r\cos\theta$，$y=r\sin\theta$.

∴Ω: $0\le z\le 9-r^2$.

令 $z=0$ ∴ $r=3$.

∴原式 $=\int_0^3 r^2dr\int_0^{2\pi}d\theta\int_0^{9-r^2}dz$

$=2\pi\int_0^3(9r^2-r^4)dr=\frac{324}{5}\pi.$

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27.(1). 球坐标: $\begin{cases} x=\rho\sin\phi\cos\theta \\ y=\rho\sin\phi\sin\theta \\ z=\rho\cos\phi \end{cases}$

积分区域为 $0\le\rho\le1$; $\frac{\pi}{2}\le\phi\le\pi$, $0\le\theta\le\frac{\pi}{2}$

$\therefore$ 原式 $=\int_0^1\int_{\frac{\pi}{2}}^{\pi}\int_0^{\frac{\pi}{2}}\rho^2\sin\phi(\rho\sin\phi\sin\theta)^2\,d\theta\,d\phi\,d\rho$

$=\int_0^1\rho^4d\rho\int_{\frac{\pi}{2}}^{\pi}\sin^3\phi\,d\phi\int_0^{\frac{\pi}{2}}\sin^2\theta\,d\theta$

$=\frac{\pi}{30}$

(6). 球1: $\rho\le R$ 球2: $\rho\le 2R\cos\phi$

$\therefore 0\le\rho\le\min\{R, 2R\cos\phi\}$, $0\le\theta\le2\pi$, $0\le\phi\le\frac{\pi}{2}$.

分界: $\phi=\frac{\pi}{3}$, $z^2=\rho^2\cos^2\phi$.

$\therefore$ 原式 $=\int_0^R\rho^4d\rho\int_0^{\frac{\pi}{3}}\cos^2\phi\sin\phi\,d\phi\int_0^{2\pi}d\theta+$

$\left(\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\left(\int_0^{2R\cos\phi}\rho^4d\rho\right)\cos^2\phi\sin\phi\,d\phi\right)\int_0^{2\pi}d\theta.$

$=\frac{7}{120}R^5\pi+\frac{1}{320}R^5\pi=\frac{59\pi}{480}R^5$

28. (2) 设 $x=r\cos\theta$, $y=r\sin\theta$, $x+y=r(\cos\theta+\sin\theta)$.

$\therefore \Omega: 1\le z\le 1+\sqrt{1-r^2}$.

xy平面上投影: $x^2+y^2=1$ $\therefore 0\le r\le 1$, $0\le\theta\le 2\pi$.

$\therefore$ 原式 $=\int_0^1 r^2dr\int_0^{2\pi}(\cos\theta+\sin\theta)d\theta\int_1^{1+\sqrt{1-r^2}}dz$

$=\int_0^1 r^2\sqrt{1-r^2}dr\int_0^{2\pi}(\cos\theta+\sin\theta)d\theta$

$=0$.

(4). 设 $x=\rho\sin\phi\cos\theta$, $y=\rho\sin\phi\sin\theta$, $z=\rho\cos\phi$.

$\therefore$ 曲面: $\begin{cases}\phi=\frac{\pi}{4}\\ \phi=\frac{\pi}{6}\\ \rho\cos\phi=1\end{cases}$

投影: 原式: $0\le\rho\le\frac{1}{\cos\phi}$, $\frac{\pi}{6}\le\phi\le\frac{\pi}{4}$, $0\le\theta\le 2\pi$.

原式 $=\iiint_\Omega(\rho+\frac{1}{\rho^2})dV$

$=\int_0^{2\pi}d\theta\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}(\int_0^{\frac{1}{\cos\phi}}\rho^2(\rho+\frac{1}{\rho^2})d\rho)\sin\phi\,d\phi)$

$=\frac{4}{3}\sqrt{2}\pi-\frac{22}{9\sqrt{3}}\pi$

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(6) 令 $u=x-a$, $v=y-b$, $w=z-c$.

$\therefore x^2+y^2+z^2=u^2+v^2+w^2+2au+2bv+2cw+a^2+b^2+c^2$.

区域: $u^2+v^2+w^2\le R^2$.

取 $u=\rho\sin\phi\cos\theta$

$v=\rho\sin\phi\sin\theta$

$w=\rho\cos\phi$ 区域为 $0\le\rho\le R$, $0\le\phi\le\pi$, $0\le\theta\le 2\pi$

$\therefore$ 原式 $=\int_0^{2\pi}d\theta\int_0^{\pi}\sin\phi\,d\phi\int_0^R\rho^4\,d\rho+\frac{4}{3}\pi R^3(a^2+b^2+c^2)$

$=\frac{4}{5}\pi R^5+\frac{4}{3}\pi R^3(a^2+b^2+c^2)$.

29. (1). 采用柱坐标.

区域: $r^2\le z\le 2-r^2$; $0\le r\le 1$; $0\le\theta\le 2\pi$.

$\therefore$ 原式 $=\int_0^1 r^4dr\int_0^{2\pi}d\theta\int_{r^2}^{2-r^2}dz$.

$=\frac{8}{35}\pi$.

(3). 采用球坐标.

区域: $0\le\rho\le 3$, $0\le\phi\le\pi$, $0\le\theta\le 2\pi$.

被积函数 $f=\rho^2\cos\phi$

$\therefore$ 原式 $=\int_0^3\rho^4d\rho\int_0^{\frac{\pi}{2}}\sin\phi\cos\phi\,d\phi\int_0^{2\pi}d\theta$

$=\frac{243}{5}\pi$

30.(1). 区域：$y^2 \le x \le 1$，$-1 \le y \le 1$，$0 \le z \le 1-x$.

$\therefore V = \int_{-1}^{1} dy \int_{y^2}^{1} dx \int_{0}^{1-x} dz$

$= \int_{-1}^{1} \left(\frac{1}{2}y^4 - y^2 + \frac{1}{2}\right) dy = \frac{8}{15}$.

(3). 取用柱坐标：$dV = r\,dr\,d\theta\,dz$.

$0 \le r \le a\cos\theta$，$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$，$-\sqrt{a^2-r^2} \le z \le \sqrt{a^2-r^2}$.

$\therefore V = \int_0^{a\cos\theta} r\,dr \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_{-\sqrt{a^2-r^2}}^{\sqrt{a^2-r^2}} dz$.

$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\int_0^{a\cos\theta} 2r\sqrt{a^2-r^2}\,dr\right) d\theta$

$= \frac{2}{9}a^3(3\pi - 4)$.

(5). 记 $A = \begin{pmatrix} 4 & 4 & 8 \\ 2 & 7 & 4 \\ 1 & 4 & 3 \end{pmatrix}$，$\det A = 20$.

$\therefore V = 20 \times 5 = 100$.

31. 区域：$-R \le x \le R$；$-\sqrt{R^2-x^2} \le y \le \sqrt{R^2-x^2}$；截面积 $S = \sqrt{3}(R^2-x^2)$.

$\therefore m = \int_{-R}^{R} \sqrt{3}(R^2-x^2)\left(1+\frac{x}{R}\right) dx = \frac{4}{3}\sqrt{3}R^3$.

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32. (3) 柱面在 x-y 平面投影为 $r=a\cos\theta$.

即: $0\le r\le a\cos\theta$, $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$.

球面上: $z=\sqrt{a^2-r^2}$ (取上半). 即 $z=\sqrt{a^2-x^2-y^2}$.

$\therefore dS=\sqrt{1+z_x^2+z_y^2}\,dxdy$.

$=\frac{a}{\sqrt{a^2-r^2}}rdr$.

$\therefore S=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^{a\cos\theta}\frac{ar}{\sqrt{a^2-r^2}}dr$.

$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}a^2(1-|\sin\theta|)d\theta=a^2(\pi-2)$.

$\therefore S_{总}=2S=2a^2(\pi-2)$.

(6). 交线在平面 $z=a$ 上.

即: $\begin{cases}z=a\\x^2+y^2=2a^2\end{cases}$.

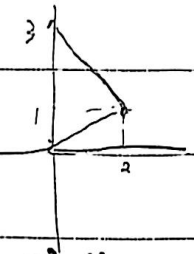
$\therefore 0\le z\le a$ 部分: $z=\frac{1}{2a}(x^2+y^2)$.

$\therefore dS=\sqrt{1+(\frac{x}{a})^2+(\frac{y}{a})^2}\,dxdy=r\frac{\sqrt{a^2+r^2}}{a}drd\theta$. $0\le r\le\sqrt{2}a$. $0\le\theta\le2\pi$

$\therefore S_1=\int_0^{\sqrt{2}a}r\frac{\sqrt{a^2+r^2}}{a}dr\int_0^{2\pi}d\theta$

$=\frac{2}{3}\pi a^2(3\sqrt{3}-1)$

33.(1)



$$m=\int_0^2\left(\int_{\frac{1}{2}x}^{3-x}(x+y)\,dy\right)dx=6.$$

质心坐标 $\bar{x}=\frac{1}{m}\int_0^2\int_{\frac{1}{2}x}^{3-x}x(x+y)\,dy\,dx=\frac{3}{4}.$

$$\bar{y}=\frac{1}{m}\int_0^2\int_{\frac{1}{2}x}^{3-x}y(x+y)\,dy\,dx=\frac{3}{2}.$$

$\therefore$ 质心为 $(\frac{3}{4},\frac{3}{2})$.

3. $0\le r\le 1+\sin\theta,\ 0\le\theta\le 2\pi$

$$\therefore m=2\int_0^{2\pi}d\theta\int_0^{1+\sin\theta}r\,dr.$$

$$=\int_0^{2\pi}(1+\sin\theta)^2\,d\theta$$

$$=3\pi.$$

$\therefore \bar{x}=0,\ \bar{y}=\frac{2}{m}\int_0^{2\pi}d\theta\int_0^{1+\sin\theta}r^2\sin\theta\,dr=\frac{5}{6}.$

$\therefore$ 质心为 $(0,\frac{5}{6})$.

34.(2) $-a\le x\le a,\ -b\sqrt{1-\frac{x^2}{a^2}}\le y\le b\sqrt{1-\frac{x^2}{a^2}}.$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}\le z\le 1.$$

$$\therefore V=\int_{-a}^{a}dx\int_{-b\sqrt{1-\frac{x^2}{a^2}}}^{b\sqrt{1-\frac{x^2}{a^2}}}dy\int_{\frac{x^2}{a^2}+\frac{y^2}{b^2}}^{1}dz.$$

$$=\int_{-a}^{a}dx\int_{-b\sqrt{1-\frac{x^2}{a^2}}}^{b\sqrt{1-\frac{x^2}{a^2}}}\left(1-\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)dy.$$ 令 $u=\frac{x}{a},\ v=\frac{y}{b}$

$$\therefore dx\,dy=ab\,du\,dv.$$

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$V = ab\int_0^1 r(1-r^2)dr\int_0^{2\pi} d\theta = \frac{1}{2}\pi ab.$

形心：$\bar{x} = \bar{y} = 0.$

$\bar{z} = \frac{1}{V} ab\int_0^1 \frac{1}{2} r(1-r^4)dr\int_0^{2\pi} d\theta = \frac{2}{3}.$

∴形心为 $(0, 0, \frac{2}{3})$.

35. 解：$\rho = kz = kr\cos\phi$

$\therefore M = \int_0^R r^2 \cdot r dr \int_0^{\frac{\pi}{2}} \sin\phi\cos\phi d\phi \cdot \int_0^{2\pi} d\theta$

$= \frac{1}{4}\pi k R^4.$

质心：$\bar{x} = \bar{y} = 0.$

$\bar{z} = \frac{k}{M}\int_0^R r^4 dr \int_0^{\frac{\pi}{2}} \sin^2\phi\cos\phi \cdot d\phi \int_0^{2\pi} d\theta = \frac{8}{15}R.$

∴质心为 $(0, 0, \frac{8}{15}R)$.